CSM
Fall 2026

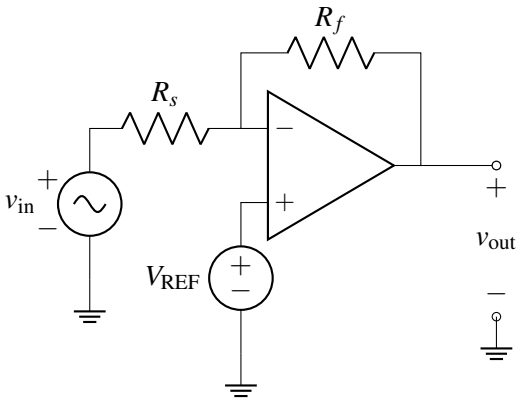
EE 64
Worksheet 9

1. Negative Feedback

Meta:

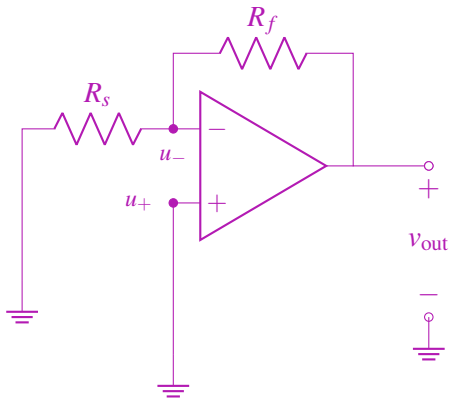
- Review the Golden Rules of Op Amps
- Explain the first Golden Rule that is applicable to every Op Amp ($I_+ = I_- = 0$) because there is no closed loop path for the current to follow. Draw out the equivalent Op Amp circuit that shows no current flowing in and the dependent voltage source to help your explanation of this Golden Rule.
- Explain how to check for negative feedback (toggling output and seeing its effects).
- Bonus: Review some applications of Op Amps to better help students understand why they are important

Are the following circuits in negative feedback? Explain your reasoning.



(a)

Solution: To check if a circuit is in negative feedback, first zero out independent voltage sources:



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Now, we need to check how the circuit responds by "wiggling the output". First, let's observe what happens when the output, v_out is increased. We see that u_- is the node inbetween a voltage divider, so

$$u_- = \frac{R_{in}}{R_{in}+R_f} v_{out}$$

Therefore, if v_{out} increases, then u_- increases as well. We know that op-amps are governed by the equation

$$v_{out} = A(u_+ - u_-)$$

where A is the internal gain of the op-amp (ideal op-amps have a gain of infinity). Make sure you understand this equation! Observe that the equivalent op-amp circuit has an internal dependent voltage source of Av_x where v_x is $u_+ - u_-$.

Now, if u_- increases, and A and u_+ remain unchanged, then $(u_+ - u_-)$ decreases, and thus $v_{out} = A(u_+ - u_-)$ decreases. When we wiggled v_{out} up, the circuit responded by decreasing v_{out}, which indicates negative feedback. Negative feedback is a good property to have! They help re-adjust to the value of the desired output when the output is too high or too low relative to the target value.

Concept check: Go through the same process, but test when v_{out} is wiggled down. Does the result of this show negative feedback as well?

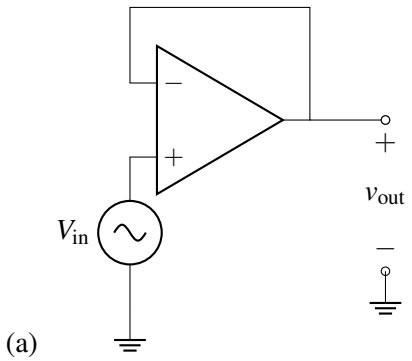
(b) Suppose that we switched the negative and positive signs on the op-amp. Is this circuit still in negative feedback?

Solution: If the terminals are swapped, we will now have a voltage divider:

$$u_+ = \frac{R_{in}}{R_{in}+R_f} v_{out}$$

When we wiggle v_{out} to go up, then u_+ will now increase. We look to the op-amp internal gain equation once again: $v_{out} = A(u_+ - u_-)$. If u_+ increases, and A and u_- remain unchanged, then $(u_+ - u_-)$ increases, and thus v_{out} increases.

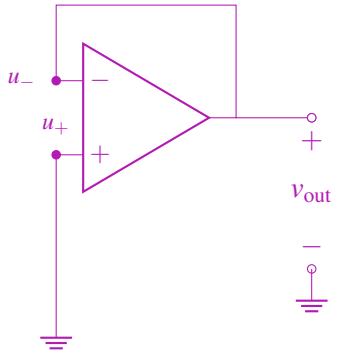
Let's observe one of the most important op-amp configurations: the unity gain buffer!



(a)

Solution: Like before, let's zero out the independent voltage source to get this circuit (recall that voltage sources become wires and current sources become open circuits):

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Like previously, let's wiggle v_{out} up first. We can observe that the negative terminal of the op-amp has the same voltage as v_{out} because they are essentially the same node. Therefore, $u_- = v_{out}$. If u_- increases, and A and u_+ remain unchanged, then $(u_+ - u_-)$ decreases, and thus $v_{out} = A(u_+ - u_-)$ decreases. We get an opposite effect: when v_{out} increases, the op=amp brings the output voltage back down.

(b) Knowing the properties of an op-amp, and that the unity gain buffer has a gain of 1, what are some possible uses of unity gain buffers?

Solution: The unity gain configuration in op-amps is extremely helpful in preventing the loading effect. If we want the output voltage of one circuit to be used as the input voltage of another, it's not as simple as putting a wire between the two. The second circuit may incur a loading effect from the first: if we just put a wire in-between, the wire will carry unwanted current into the second circuit and will change some of its properties. This is where op-amps come in! Op-amp terminals don't allow current to go in (recall the first Golden Rule: $I_- = I_+ = 0$, and thus, v_{out} of the op-amp will give us the wanted voltage, without any of the current from the first circuit. Because this is the *unity gain* buffer, $v_{out} = v_{in}$, and the desired voltage will be preserved completely.

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2. Voltage Summers

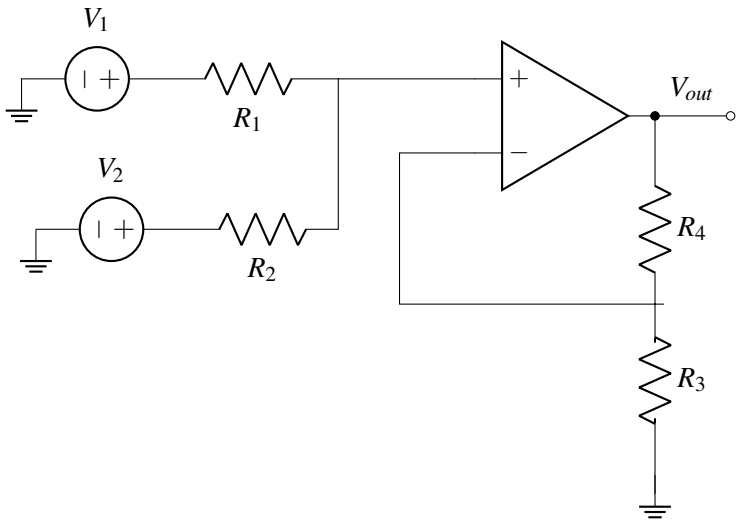
Learning Goal: This problem uses basic circuit analysis techniques to find the response of a summer circuit.

Relevant Notes: *Note 19* goes different op-amp circuit topology and corresponding derivations.

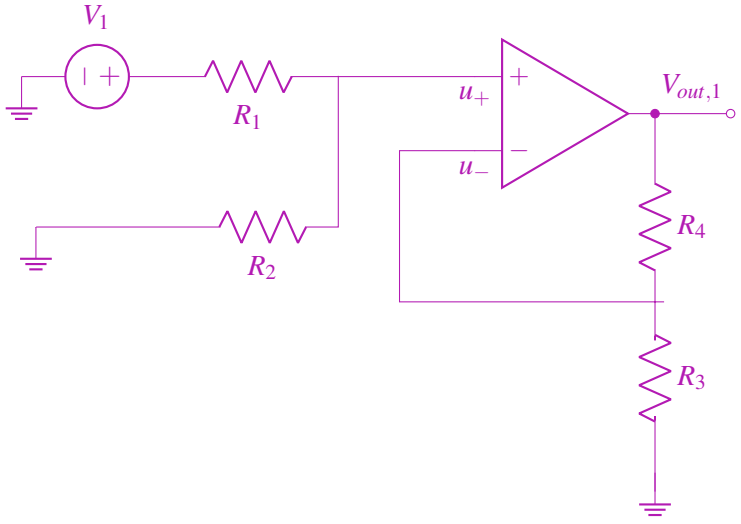
Meta:

- Mention/ Show that this derivation could have been done using KCL instead of voltage divider equation.
- Highlight that we could use voltage divider equation because the input currents are zero, i.e. $I_+ = 0$ and $I_- = 0$.
- Also highlight that negative feedback needs to be ensured before using the second golden rule.
- Show that this circuit can be pattern matched with the **voltage summer** and the **non-inverting amplifier** circuits from the first page of the worksheet. Those equations can be utilized to find the final response too.

(a) Calculate V_{out} in terms of V_1 and V_2. Assume that $R_1 = R_2$. Use superposition.



Solution: Let's first consider the case when only V_1 is active. We deactivate voltage source V_2 by replacing it with a wire:



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We see that u_+ is the middle node of a voltage divider with resistors R_1 and R_2. We also know that input current $I_+ = 0$. So we can find u_+ using:

$$u_+ = \frac{R_2}{R_1 + R_2} V_1.$$

Similarly u_- is the middle node of a voltage divider with resistors R_3 and R_4. We know that input current $I_- = 0$. We can find u_- as:

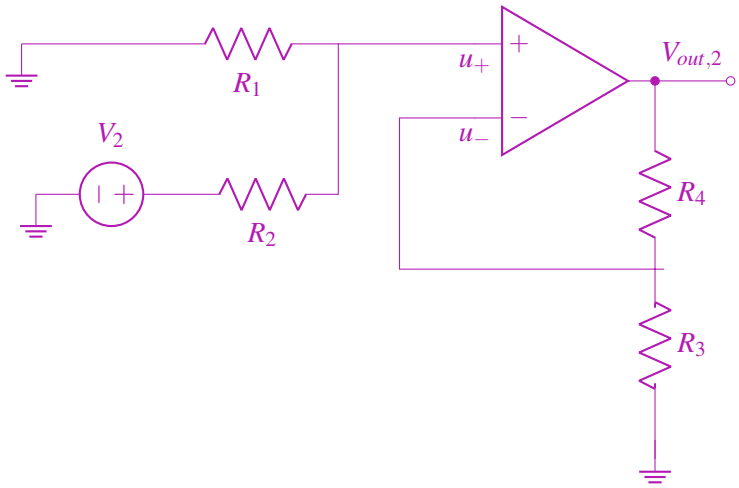
$$u_- = \frac{R_3}{R_3 + R_4} V_{out,1}.$$

Now we have to check if the circuit is in negative feedback. If we move the negative input of the op amp u_- upward, $V_{out,1} = A(u_+ - u_-)$ moves downward and as a result $u_- = \frac{R_3}{R_3+R_4} V_{out,1}$ moves downward. So the result of the initial stimulus goes in the opposite direction of the initial stimulus, which is the requirement for negative feedback.

Since the circuit is in negative feedback, we can apply the second golden rule:

$$\begin{aligned} u_+ &= u_- \\ \implies \frac{R_2}{R_1 + R_2} V_1 &= \frac{R_3}{R_3 + R_4} V_{out,1} \\ \implies V_{out,1} &= \left(1 + \frac{R_4}{R_3}\right) \frac{R_2}{R_1 + R_2} V_1 \end{aligned}$$

Next we consider the case when only V_2 is active. We deactivate voltage source V_1 by replacing it with a wire:



Following a similar process we have:

$$\begin{aligned} u_+ &= \frac{R_1}{R_1 + R_2} V_2; \\ u_- &= \frac{R_3}{R_3 + R_4} V_{out,2}. \end{aligned}$$

(Note that u_+ is now the voltage across R_1, while for the previous scenario u_+ was the voltage across R_2.)

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Using the second golden rule we have:

$$u_+ = u_-$$
$$\implies \frac{R_1}{R_1+R_2}V_2 = \frac{R_3}{R_3+R_4}V_{out,2}$$
$$\implies V_{out,2} = \left(1+\frac{R_4}{R_3}\right)\frac{R_1}{R_1+R_2}V_2$$

Now using the superposition theorem, we can find V_{out}:

$$V_{out} = V_{out,1} + V_{out,2}$$
$$\implies V_{out} = \left(1+\frac{R_4}{R_3}\right)\frac{R_2}{R_1+R_2}V_1 + \left(1+\frac{R_4}{R_3}\right)\frac{R_1}{R_1+R_2}V_2$$

Note that a similar expression can be reached by using the equations for the voltage summer and the non-inverting amplifier. Pattern matching this circuit with the voltage summer and the non-inverting amplifier and then employing the given equations is a valid technique to solve this problem.

(b) What values should we select for R_1, R_2, R_3, and R_4 such that $V_{out} = V_1 + 2V_2$?

Solution: From the last part, we have

$$V_{out} = \left(1+\frac{R_4}{R_3}\right)\frac{R_2}{R_1+R_2}V_1 + \left(1+\frac{R_4}{R_3}\right)\frac{R_1}{R_1+R_2}V_2$$

Comparing with $V_{out} = V_1 + 2V_2$, we have:

$$\left(1+\frac{R_4}{R_3}\right)\frac{R_2}{R_1+R_2} = 1 \tag{1}$$
$$\left(1+\frac{R_4}{R_3}\right)\frac{R_1}{R_1+R_2} = 2 \tag{2}$$

Dividing equation (1) with equation (2) we get:

$$\frac{R_2}{R_1} = \frac{1}{2}$$
$$R_1 = 2R_2$$

Now if we choose $R_2 = 100\Omega$ and $R_1 = 200\Omega$, we have

$$\frac{R_2}{R_1+R_2} = \frac{100}{200+100} = \frac{1}{3} \tag{3}$$
$$\frac{R_1}{R_1+R_2} = \frac{200}{200+100} = \frac{2}{3} \tag{4}$$

Plugging in the values of equation (3) in equation (1) (or equation (4) in equation (2)), we have:

$$\left(1+\frac{R_4}{R_3}\right)\frac{1}{3} = 1$$
$$1+\frac{R_4}{R_3} = 3$$
$$\frac{R_4}{R_3} = 2$$
$$R_4 = 2R_3$$

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We can choose $R_3 = 100\Omega$ and $R_4 = 200\Omega$, so that the output voltage is:

$$V_{out} = \left(1+\frac{200}{100}\right)\frac{100}{200+100}V_1 + \left(1+\frac{200}{100}\right)\frac{200}{200+100}V_2 = (1+2)\frac{1}{3}V_1 + (1+2)\frac{2}{3}V_2 = V_1 + 2V_2.$$

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3. Multi-stage Amplifier

Learning Goal: The objective of this problem is to understand how multiple stages of op-amp circuits can be used to achieve a specific circuit gain.

Relevant Notes: *Note 19 Section 19.5* goes over inverting and non-inverting amplifiers.

Meta: Highlight that it is not necessary to have a buffer between two inverting amplifiers. Although resistors R_3 and R_4 are drawing current from the first amplifier output, the first output $V_{o,1}$ is not going to change. This is because the output of an ideal op-amp can supply any current to the load without changing the output voltage. So in this case there will be no loading effect.

Mention that loading effect will NOT occur if the second amplifier is not drawing any input current OR the first amplifier output voltage does not change.

(a) What is the range of values that we can scale V_{in} by when using a non-inverting op amp? (What are possible values for the gain?)

Solution: Recall that when using a non-inverting op amp, the equation for V_{out} is given by

$$V_{out} = V_{in}(1 + \frac{R_2}{R_1})$$

where R_2 is the upper resistor and R_1 is the lower one (connected to ground). The circuit gain G is represented by

$$G = \frac{V_{out}}{V_{in}} = 1 + \frac{R_2}{R_1}$$

We can choose any values for resistance from [0, ∞); then, the minimum gain would be 1 if we chose $R_2 = 0$; the maximum gain approaches infinity as we choose some very large R_2 and/or a very small R_1. Hence, the range of gains is from 1 to infinity, and our range of values for V_{out} is $[V_{in}, \infty)$.

(b) What is the range of values that we can scale V_{in} by when using an inverting op amp? (What are the possible values for the gain?)

Solution: Recall that when using an inverting op amp, the equation for V_{out} is given by

$$V_{out} = -V_{in}\frac{R_2}{R_1}$$

. The circuit gain is represented by

$$G = -\frac{R_2}{R_1}$$

.

Again, we can choose any values for resistance from [0, ∞); then, the minimum absolute value scaling would be 0 if we chose $R_2 = 0$; the maximum absolute value scaling approaches infinity as we choose some very large R_2 and/or a very small R_1. Hence, the range of gains is from 0 to -∞, and our range of values for V_{out} is also $(-\infty, 0]$.

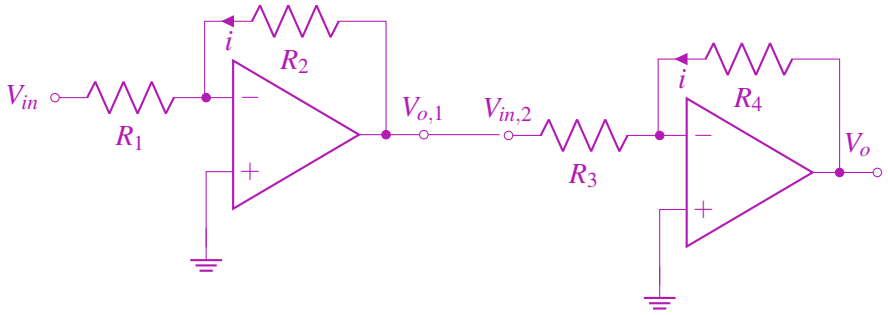
(c) Can you design a single inverting/ non-inverting amplifier with circuit gain $G = 0.5$? If not, what range of gain values is not reachable using a single inverting op amp or a single non-inverting op amp?

Solution: From part (a), we found that a non-inverting op amp can only reach gain values from [1, ∞). From part (b), we found that an inverting op amp can only reach gain values from (-∞, 0]. So using only a single op-amp, we cannot reach values between (0, 1), specifically $G = 0.5$.

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(d) How would you construct a circuit using inverting/ non-inverting amplifiers so that the overall circuit gain is $G = 0.5$?

Solution: We can use two inverting op-amps to achieve overall gain $G = 0.5$. Say the first op-amp has gain G_1 and the second op-amp G_2, so $G = G_1 G_2$. From part (b), these gains can take on values from (-∞, 0], so multiplying them together helps reach $G = 0.5$.



For the first inverting op-amp, we have $G_1 = -\frac{R_2}{R_1}$, and for the second, $G_2 = -\frac{R_4}{R_3}$. Plugging these into the equation for G:

$$G = G_1 G_2 = \frac{R_2 R_4}{R_1 R_3} = 0.5$$

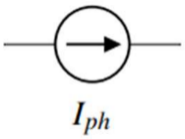
$$R_1 R_3 = 2 R_2 R_4$$

To reach out desired gain, we can pick any combination of R_1, R_2, R_3, and R_4 that satisfies this equation. One such solution is $R_1 = 200\Omega$ and $R_2 = R_3 = R_4 = 100\Omega$.

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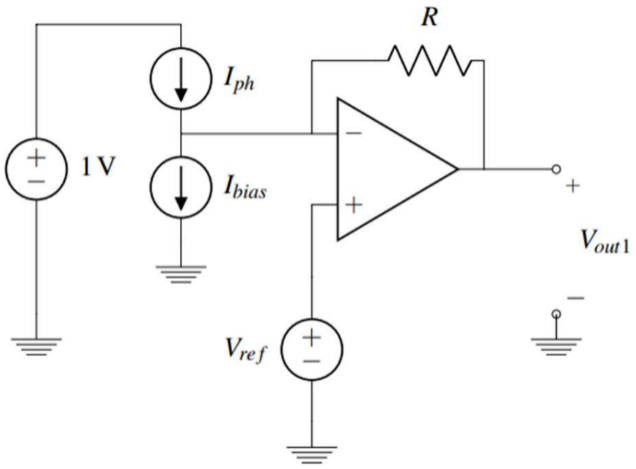
4. Op-Amp Analysis

After finishing EE64, you get the opportunity to work with Professor Vladimir. His group designs computer chips that communicate with light, and your job is to design the receiver. Your receiver connects to a photodetector, that can be modeled as current source.



When light is shining on the receiver, the current becomes $I_{ph} = 1\mu A$. Otherwise, with no light shining on the receiver the current is $I_{ph} = 0A$. The circuit should output $0V$ to the processor when there is light shining on the receiver, and output $1V$ to the processor when there is no light shining on the receiver.

(a) Your friend suggests the following circuit to turn the current into a voltage. Find the output voltage, V_{out} as a function of I_{ph}, and find a value of R such that the circuit outputs $0V$ when $I_{ph} = 1\mu A$. Here, $I_{bias} = 0.5\mu A$ and $V_{ref} = 0.5V$. Show your work.



Meta: This question comes from Q9(a) of Spring 2019's Midterm 2.

Solution: To form an expression of V_{out}, we would need to solve this circuit. Therefore, let us first come up with a series of passive sign convention to maintain during the analysis. In the solution, i_R is defined to flow towards the negative node of op-amp.

We can observe that $V_out - i_R R = V_-$. If we increase V_out, V_- will increase and thus decrease the value of $V_+ - V_-$, leading to a decrease in V_out. An increase in output leads to a decrease in output, therefore, negative feedback exists.

We may apply the golden rules $V_+ = V_-$ and $i_+ = i_- = 0A$. In particular, the first described equation brings up $V_+ = V_- = V_{ref}$, while the second description facilitates the KCL equation at the negative terminal's node:

$$\begin{aligned} i_in &= (I_{ph} - I_{bias}) + i_R \\ &= 0 = i_{out} \end{aligned}$$

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Now that we have these information, let's come up with an expression for V_{out} in terms of I_{ph}:

$$\begin{aligned} V_{out} &= V_- + i_R R \\ &= V_{ref} - (I_{ph} - I_{bias})R \end{aligned}$$

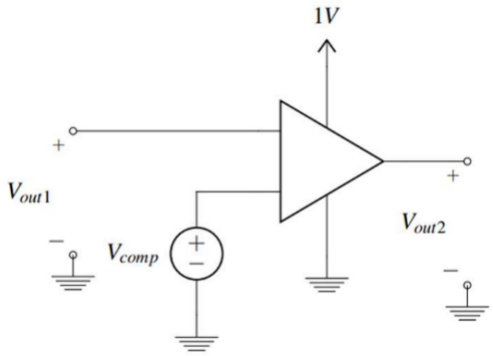
For outputting $0V$, the resistor must have a resistance value R_0 such that:

$$V_{ref} - (I_{ph} - I_{bias})R_0 = V_{out} = 0V$$

In analysis, at the end of algebra, the answer awaits:

$$R_0 = \frac{V_{out} - V_{ref}}{I_{bias} - I_{ph}} = 1M\Omega$$

(b) After building the circuit and testing it, you find that I_{ph} varies as you get closer or further from light. This causes the output voltage of your circuit to not be precisely $0V$ or $1V$.
However, the processor needs exactly $0V$ when light is shining, and $1V$ when no light is shining. Vlad advises you add a comparator to your design. Let $I_{ph,light}$ denote the minimum current indicating there is enough light.
Finish the circuit below by calculating the value of V_{comp} for the comparator. Also, select the polarity of the comparator by drawing a '+' or '-' in corresponding boxes in the diagram below.



Meta: This question comes from Q9(b) of Spring 2019's Midterm 2.

Solution: In a comparator of similar setup:

$$V_{out} = \begin{cases} V_{DD} & V_+ > V_- \\ V_{SS} & V_+ < V_- \end{cases}$$

In the context of the prompt, $V_{DD} = 1V$ and $V_{SS} = 0V$.
Meanwhile, the prompt demands us to produce a comparator setup such that:

$$V_{out} = \begin{cases} 1V & V_{out,1} > V_{comp} \\ 0V & V_{out,1} < V_{comp} \end{cases}$$

Therefore, the upper terminal connecting to $V_{out,1}$ should be positive '+', and the lower terminal connecting to V_{comp} should be negative '-'.
Next we will determine V_{comp}.
According to the prompt: "The circuit should output $0V$ to the processor when there is light shining on

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the receiver, and output $1V$ to the processor when there is no light shining on the receiver". The trend is then that the higher $V_{out,1}$, the less light there is. Therefore, the reference voltage of comparator, V_{comp}, should be between the maximum voltage output when there is light (maximum $V_{out,1}$ recognizable as having light) and the maximum possible voltage output (1V).
Let $I_{ph,light}$ denote the minimum current indicating there is enough light, then:

$$V_{ref} - (I_{ph,light} - I_{bias})R_0 < V_{comp} < 1V$$

such that for any situation such that the output voltage is smaller than $V_{ref} - (I_{ph,light} - I_{bias})R_0$, it is recognized that light exists.
The ideal voltage for V_{comp} to be would be the midpoint of range of all possible values; thus:

$$\begin{aligned} V_{comp} &= \frac{V_{ref} - (I_{ph,light} - I_{bias})R + 1V}{2} \\ &= \frac{1.5V - (I_{ph,light} - I_{bias})R}{2} \end{aligned}$$

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